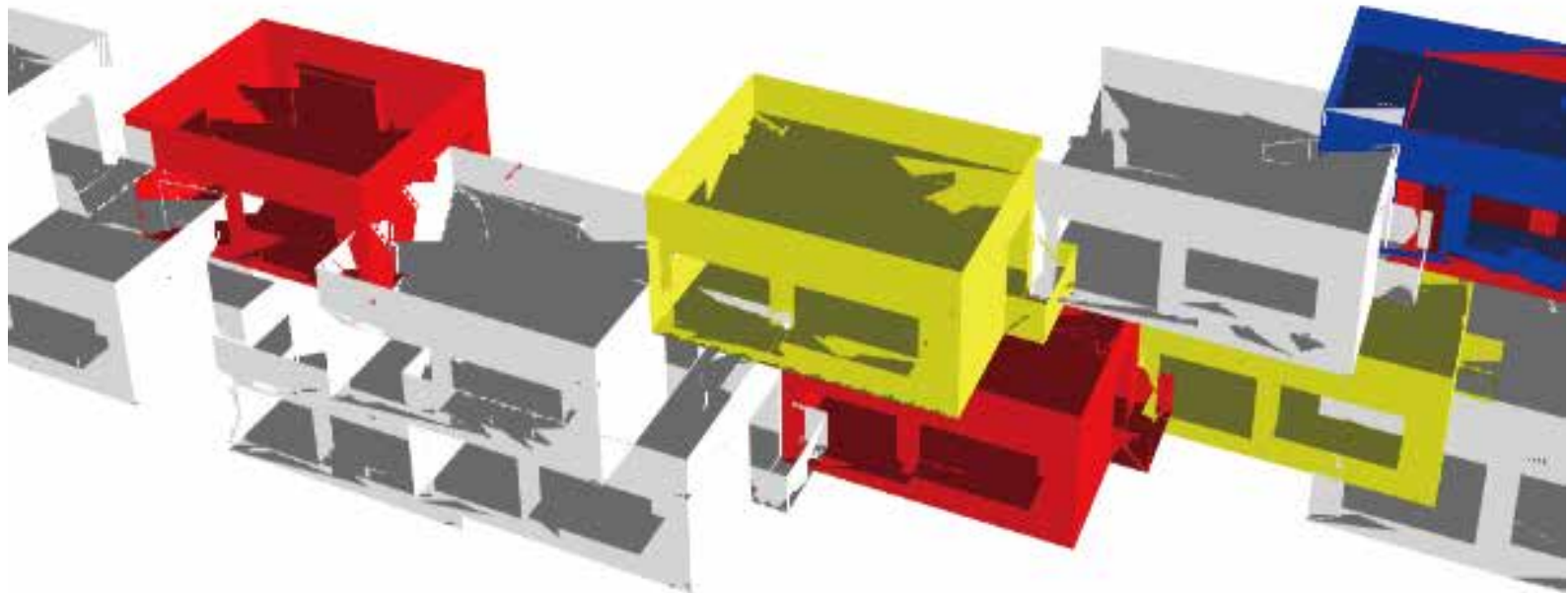


WHO DESIGNED THE SPACE?

Negotiate the algorithm to design the building

An architectural design simulator that recreates the design process within the context of the "collective practice" era.

Examples of work processes and references



Observation Report Who designed the space?		
HUMAN AUTHORSHIP 50%	AI AUTHORSHIP 50%	Data
		TRIM 2 UNDO 0 REMOVED 2
This is an architectural design with a equal negotiation.		
Wow, an equal division of labor! Human-machine collaboration makes the work effortless!(*◡◡*)		

Background research&Prototyping

1.1-See a fun topic

I saw an article about **designer** in *Age of Collective Practice*

"The question is no longer whether machines assist (designer) in design, but how **much they participate in it**; how architects **should position themselves** within this **expanding ecology of human and non-human contributors**."

—<The Machine in the Age of Collective Practice> 1

Its the issue that related to **Designers in the various field**

Q1-Who designed which part?

Q2-How much did they do?

Q3-Has the position of a designer changed in such a background?

1.2-What I want to test through design

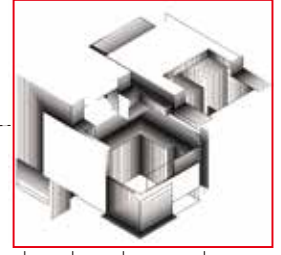


architectural design topic



CA algorithm tech

my own outcome



digital media visualization

Thats quite a topic to be discussed,i have the oppotunity to think&research&give my own answer.

1.3-Further research

searching for some theories to form my own answer to these 3 questions

Q1&Q2

Tools are not neutral.They **structure what can be designed** before design begins. 2

Machines do **not only produce forms**.They **change the conditions** under which forms **are imagined**. 3

Architectural tools do **not simply execute decisions**.They structure what can be imagined, represented and decided. 4

A1&A2:

Algorithm tech&AI are like other tools to provide a new platform to **explore humans creation&production**. While human should imagine and explore new possibilities based on new platform.

Q3

The architect no longer only produces a fixed form, but 5 **defines a system of rules** through which forms can emerge.

Creativity, likewise, is less a solitary act of invention 6 than a **negotiation within systems** of shared intelligence.

If the algorithm's creator occupies the core position of creativity, then sexuality has not disappeared but has been **reorganized**. 7

A3:Designers(eg,architects) still play a role as a **negotiator** that negotiate with algorithm tech by defining new algorithmic rules,**new algorithmic** systems, new algorithmic shapes,functions,etc ,and **take the responsibility** of these parts during such an age of collective practice.

Conclusion:

Starting from the question of the **collaborative division of labor between designers and algorithms** in the design process of collective practice era

this research inspired me to frame the project as a **collective practice** that can **simulating the negotiation process between contemporary designers and algorithmic technologies**,in order to discuss this question.

1.4-Decide the form of this collective practice project

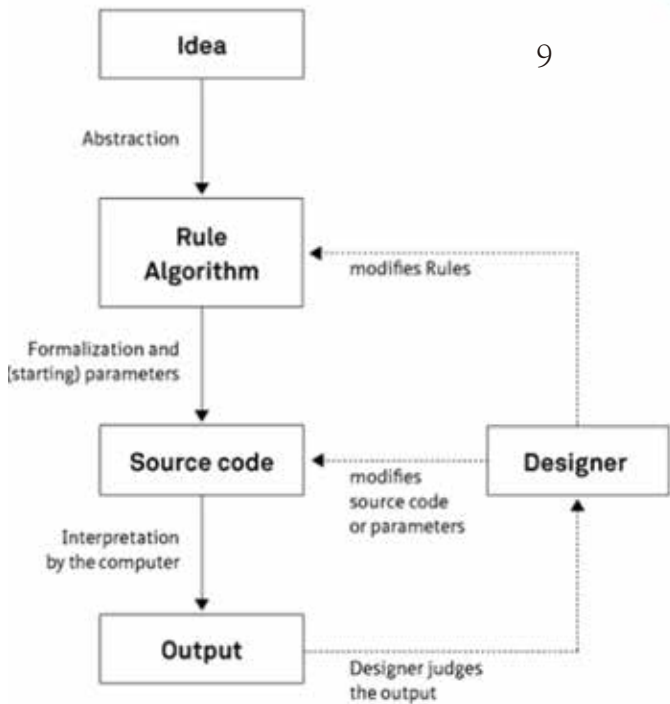
original sketch

After long search, I found a Grasshopper component set that allows the user to define the growth logic themselves, which sparked my initial sketch of the project.

At first, I looked at a lot of reference ideas but couldn't find anything I was satisfied with — it was driving me crazy

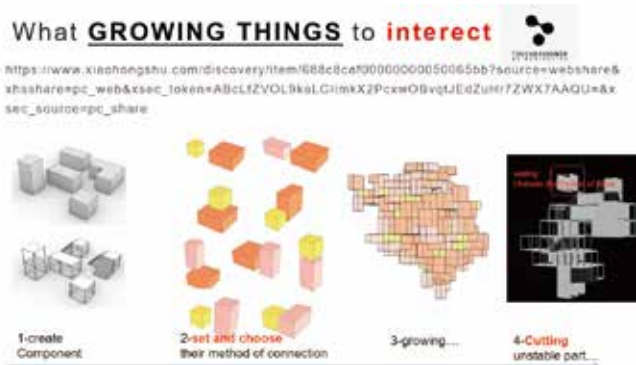
1.5-Deciding the Interaction elements & Technical frame

9

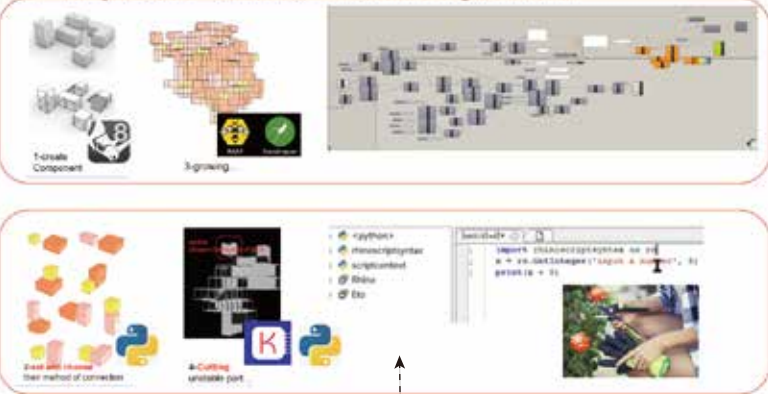


Copyright Hartmut Bohnacker, Julia Laub, Benedikt Groß, Claudius Lazzaroni (2009)
Book „Generative Gestaltung“, www.generative-gestaltung.de

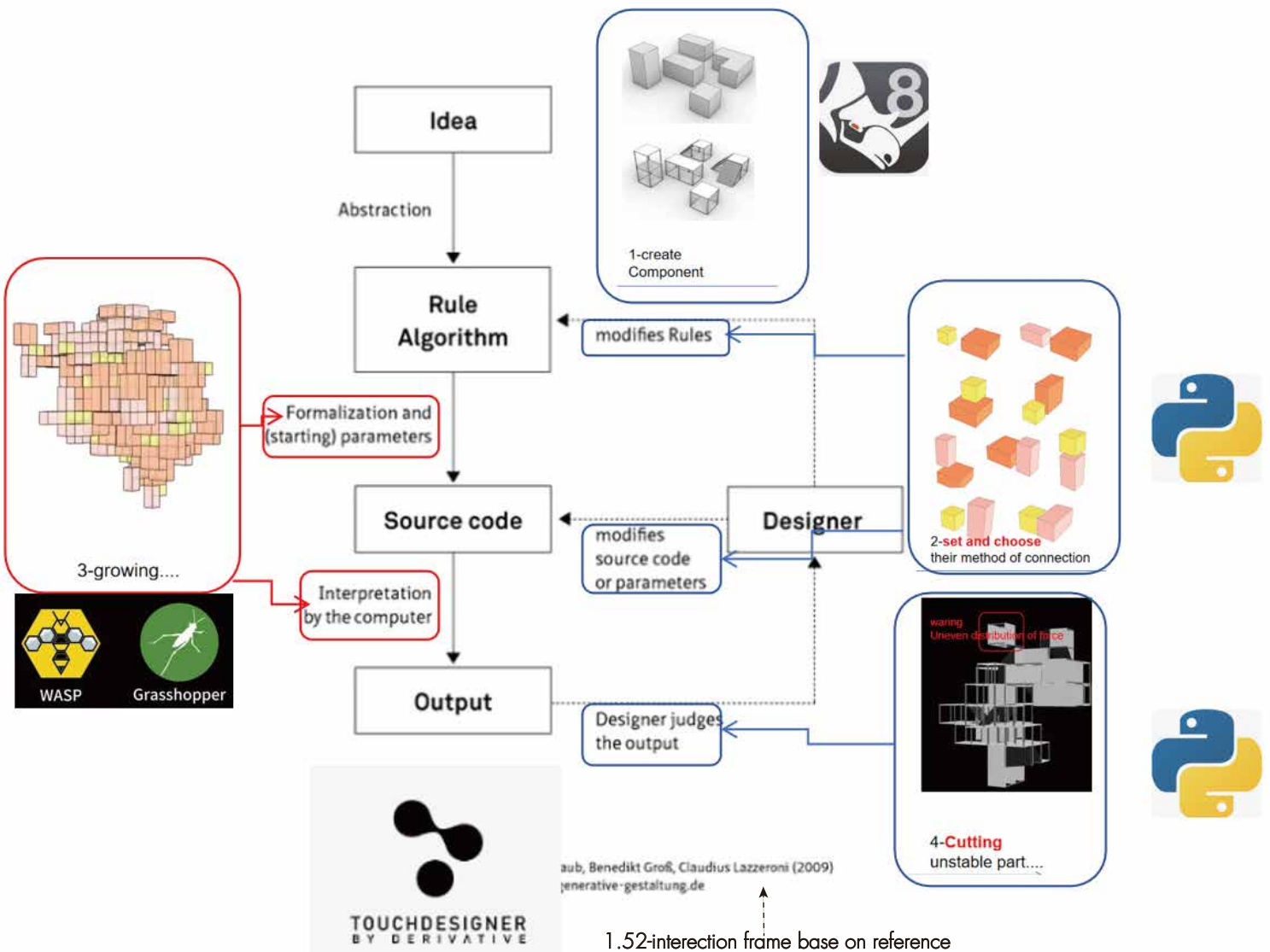
1.51-exploring the process by which contemporary designers negotiate with algorithms to create designs, in order to refine my own interaction process.



S1-key tech--Slime Mould Algorithm



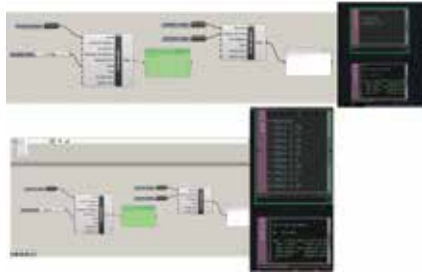
1.52-Seeking an architectural growth algorithm to serve as the primary growth logic.(Rhino GrassHopper)



1.52-interction frame base on reference

Technique Experiments

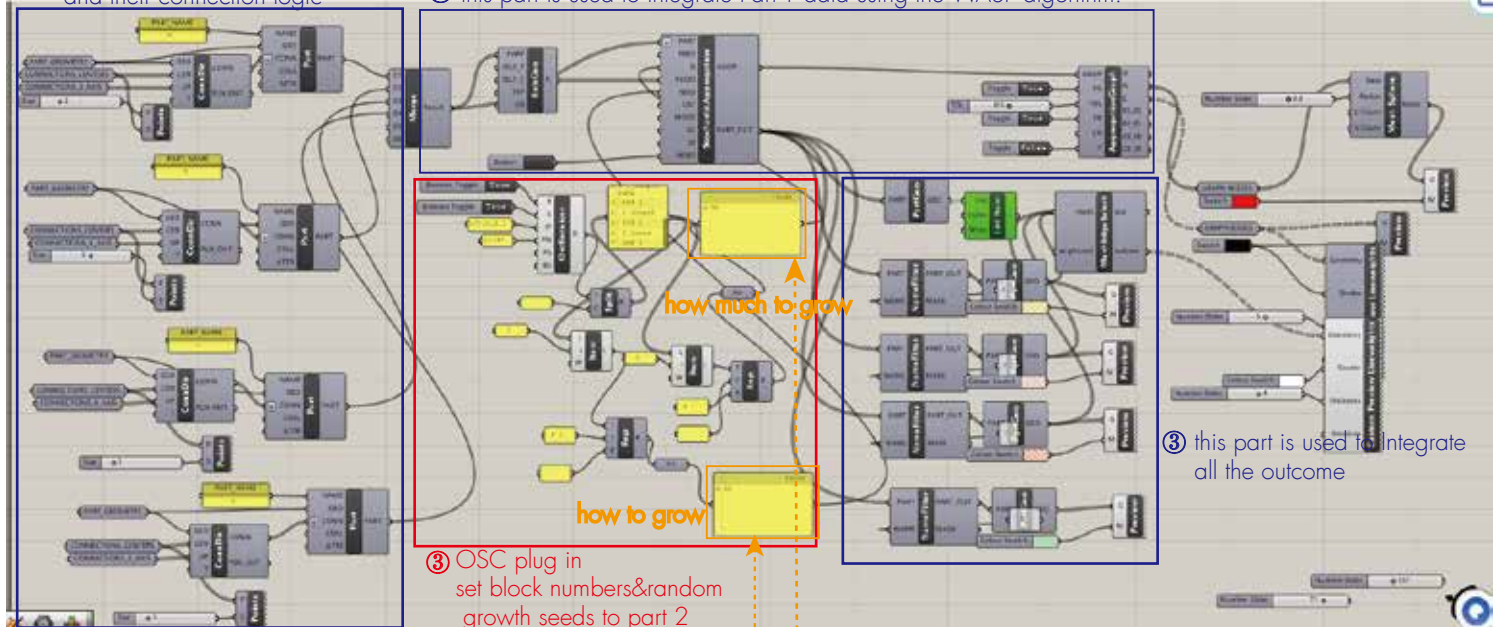
2.1-Rhino Grasshopper-WASP+TD constant&OSC DATA



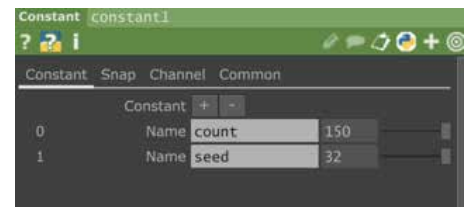
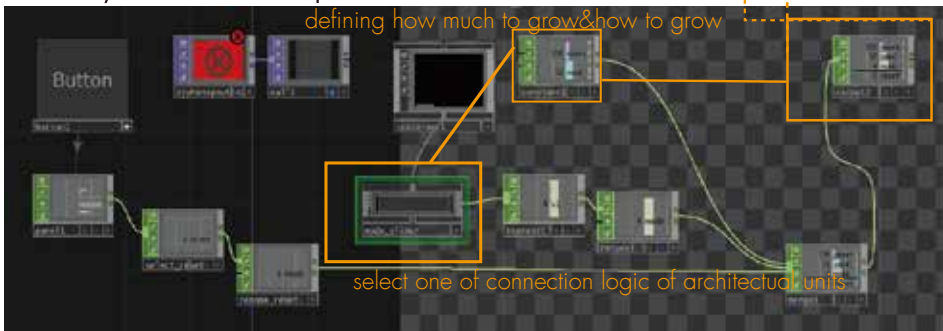
I found a **GrassHopper Plug-in** that can directly receive data from TD's OSC DAT & Constant Chop

① this part is used to set units and their connection logic

② this part is used to Integrate Part-1 data using the WASP algorithm.



Add the Gh OSC plug-in into Gh growth algorithm to be my basic frame of GH part.



Conclusion:
Now i've proved that using TD to control the growth of the virtual blocks in GH is **accessible**.

But so far the system **lack** the growing animation&triming interection&score board and OBS is not suitable for showing blocks.

I need to make more great modifications to Enrich its visual and interactive experiences.



Try visualizing the components that control the building's growth within TD.

3.1-Big modification 1 setting TD based visualization&growing animation

	0	1	2	3	4	5	6	7	8	9	10	11	12
row	id	type	x	y	z	u	v	w	x	y	z	stable	removed
0	0	A	10607.61784110001	15267.5448215512424	2393.38883100470087	0	0	0	0	0	0	1	0
1	0	A	4515.4088181470344	14120.802746093746	302.4375	0	0	0	0	0	0	1	0
2	0	A	7932.5090947490355	15551.810167819757	3537.7623447777480	0	0	0	0	0	0	1	0
3	0	A	3267.17197862859	9792.51354607005	6109.151977123863	0	0	0	0	0	0	1	0
4	0	A	10607.617830344671	880.511905107818	181.27228812814545	0	0	0	0	0	0	1	0
5	0	A	3267.1719628595047	6184.362888701624	1892.388881270218	0	0	0	0	0	0	1	0
6	0	A	4071.2738958488418	888.5119517543808	181.37272784162694	0	0	0	0	0	0	1	0
7	0	A	6728.382835709318	7115.440618180647	1424.0207817460232	0	0	0	0	0	0	1	0
8	0	A	4073.2738126352485	1845.9252778177403	5182.280480055118	0	0	0	0	0	0	1	0
9	0	A	4073.27387785214	15282.344815325709	3552.3944404096032	0	0	0	0	0	0	1	0
10	0	A	8165.48291015625	14118.802746093746	6324.21159140625	0	0	0	0	0	0	1	0
11	0	A	8165.48291015625	6421.64913404237	362.4374530959883	0	0	0	0	0	0	1	0



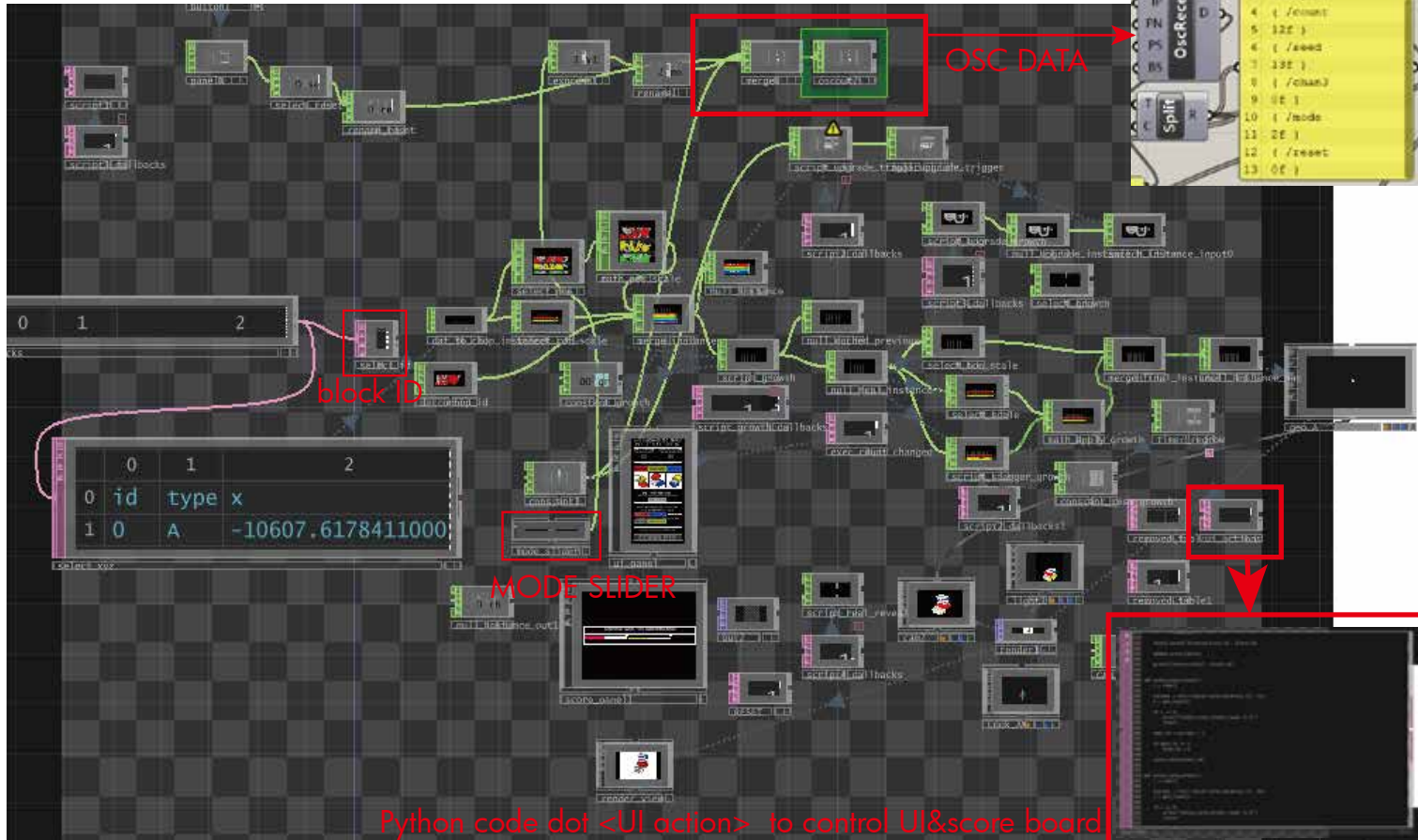
dynamically updated
growth CSV file from
Rhino Grasshopper

[illegible]

Final version of interactive workflow

3.2-Big modification 2

UI&score board



WHO DESIGNED THE SPACE?

Negotiate with the algorithm to design the building
control the OSC DATA to define how much to grow & how to grow

STEP 1 | UNIT COUNT

How many units to grow?

12

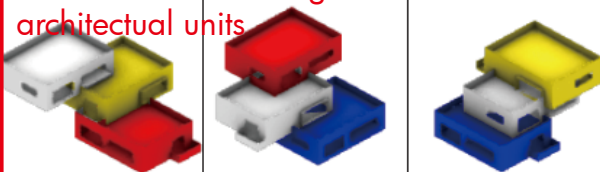
STEP 2 | RANDOM SEED

How many branches to grow?

13

STEP 3 | UNIT CONNECTION MODE

control mode slider to select one of connection logic of architectural units



STEP 4 | START GENERATING

[Double click] to [start] generating and see how it grows...

Start

STEP 5 | PRUNE & MODIFY THE STRUCTURE

[Double click] NEXT / PREV TO select a block.

PREV

NEXT

SELECTED: 0

[Double click] [TRIM] to remove it/[CANCEL] to undo trimming

TRIM

CANCEL

[Double click] [COMPLETE] if you finished pruning

COMPLETE

Observation Report | Who designed the space?

HUMAN AUTHORSHIP 50% AI AUTHORSHIP 50% Data

TRIM 2 UNDO 0 REMOVED 0

This is an architectural design with a equal negotiation.

Wow, an equal division of labor! Human-machine collaboration makes the work effortless! (***)

scoring rules and concluding remarks

Scoring is primarily based on the proportion of design authorship by the player versus those determined by the algorithm.

Performing actions STP1 through STP3 awards 10 points per action (considered basic player inputs).

Performing action STP5 awards 15 points per action (considered an advanced maneuver, hence the higher score).

However, since the WASP algorithm is integral to the game's design, the player's maximum possible score is 70 points; a full mark is unattainable. This demonstrates that even when aided by algorithms, designers must still bear the corresponding responsibilities in their work.

block ID

```
def select_block(block_id):
    try:
        block_id = int(block_id)
    except:
        print("[select_block] invalid")
        return
    c = get_count()
    if c > 0:
        if block_id < 0:
            block_id = 0
        if block_id >= c:
            block_id = c - 1
    root().store('selected_block_id', block_id)
    update_score_table()
    print("[select_block]", block_id)

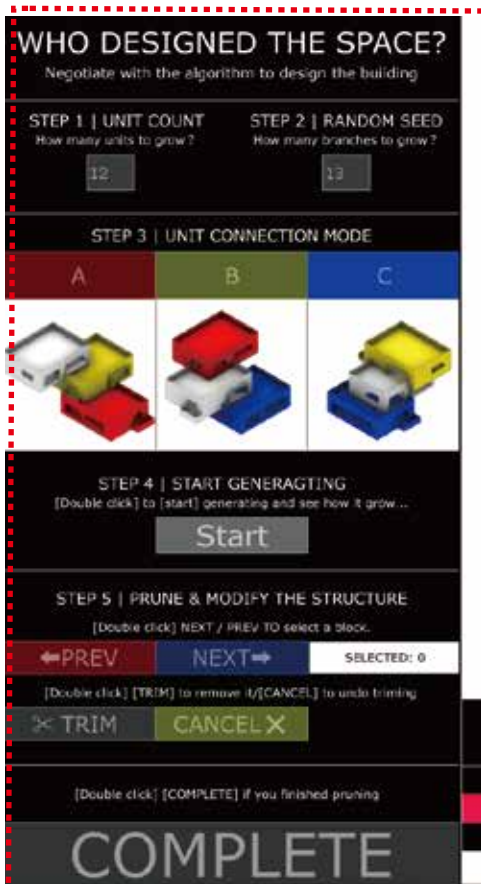
def select_next_block():
    r = root()
    current = int(r.fetch('selected_block_id', 0))
    c = get_count()
    if c <= 0:
        print("[select_next_block] count is 0")
        return

    def undo_last_trim():
        r = root()
        removed = get_removed_ids()
        if len(removed) > 0:
            print("[undo_last_trim] nothing to undo")
            return
        last = removed.pop()
        set_removed_ids(removed)
        undo_count = int(r.fetch('undo_action_count', 0))
        r.store('undo_action_count', undo_count + 1)
        r.store('selected_block_id', last)
        update_removed_table()
        calculate_score()
        update_score_table()
        print("[undo_last_trim] undo:", last)

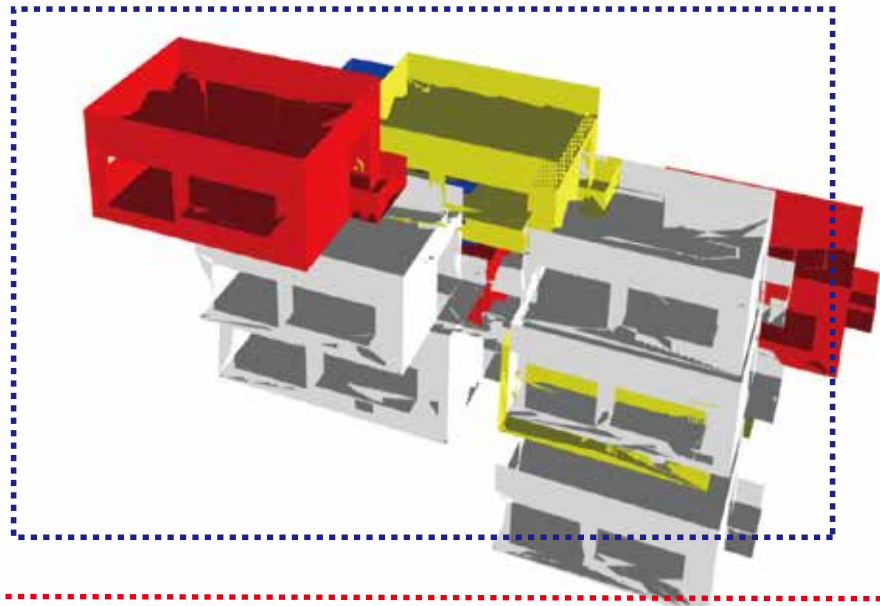
    set_removed_ids([])
    update_removed_table()
    calculate_score()
    update_score_table()
    print("[clear_trim] cleared")
```

Code for selecting & trimming block units

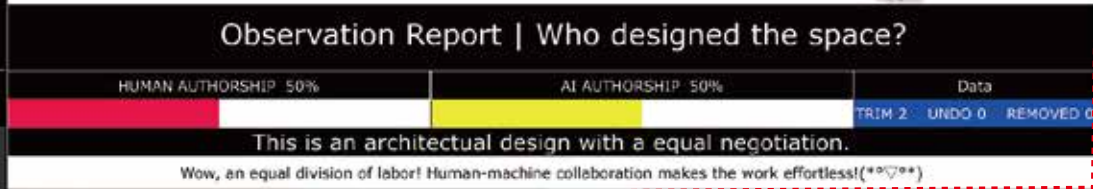
Visual Effects Design



Interface Reference



architectural unit Reference



11 Composition VII (The Three Graces)
Theo van Doesburg



12 Maison Particulière (Private Residence)
Theo van Doesburg&Cornelis van Eesteren

Why did I choose De Stijl(neoplasticism)-style graphic and architectural design as the primary visual language?

1-With a minimalist form, blocky, cubic geometry, and a rich color palette, it conveys a strong sense of rational design.

I believe this rational aesthetic aligns perfectly with my design theme:

1- First, it embodies a modernist architectural style that creates a compelling visual effect of "growth" when combined with Grasshopper (reminiscent of LEGO).

2- Many works of Neoplasticism emerged during the Second Industrial Revolution, representing an avant-garde design movement driven by the era's emerging technologies (machines). This parallels the current trend where algorithmic design is spearheading a new design movement—characterized by a blend of rationality and rules that nonetheless allows for rich, dynamic variation.